OpenWebUI on Azure

A repeatable, closed-environment foundation for client chatbots

Architecture · Infrastructure · Delivery · Ways of working

~15 min + Q&A

1 · Framing

The ask: a generic chatbot foundation, deployable per client, that becomes the base for their AI initiatives.

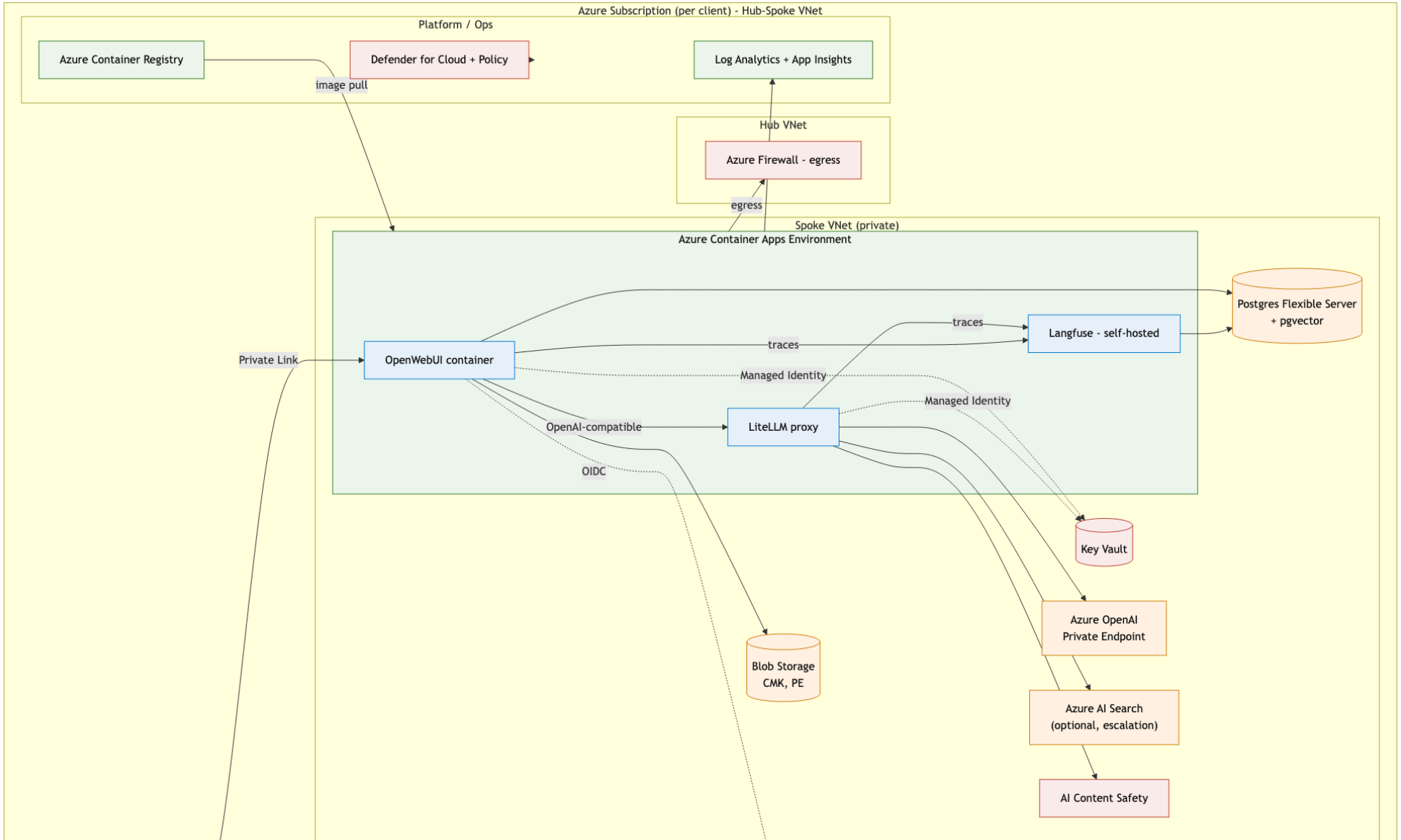
Constraints:

- Runs entirely on Azure — closed, secure data plane
- Infra + app fully scripted
- CI/CD for infra changes, new models, and bug fixes

My design stance:

- Boring defaults, platform-native services
- One template → many clients, parameterised not forked
- Optimise for *iteration speed on prompts and models*, not infra heroics

2 · Target architecture



2 · Architecture — component choices

Layer	Choice	Why this, not that
Ingress	**Front Door + WAF**	TLS, OWASP, DDoS; private origin
Identity	**Entra ID OIDC** + Managed Identities	SSO per client tenant, no app secrets
Compute	**Azure Container Apps** (internal, VNet)	Right-sized; revisions + canary; no K8s tax
State DB	**Postgres Flexible Server** (private)	OpenWebUI first-class support
Vectors	**pgvector** → escalate to **Azure AI Search**	Start simple; upgrade when retrieval demands
LLM	**Azure OpenAI** + **LiteLLM** proxy	One endpoint for OpenWebUI, many models behind it
Files	**Blob Storage** (private endpoint, CMK)	RAG sources + uploads
Secrets	**Key Vault** + Managed Identity	Zero secrets in pipelines/app
Obs (infra)	**Log Analytics + App Insights**	Native, alert rules as code
Obs (LLM)	**Langfuse** (self-hosted Container App)	Traces, token cost, eval gates
Network	Hub-spoke VNet, Private Endpoints, **Azure Firewall** egress	Satisfies "closed environment"

2 · Architecture — key trade-offs

Container Apps vs AKS

- AKS only if the client already runs K8s. Otherwise Container Apps wins on ops cost and time-to-first-deploy.

pgvector vs Azure AI Search

- pgvector: one less service, good enough for most corpora.
- AI Search: hybrid BM25+vector, semantic ranker, skillsets. Worth the jump when retrieval quality becomes the bottleneck.

Single-tenant per client vs shared multi-tenant

- Recommend **per-client stack**. Data isolation, blast-radius control, per-client compliance posture. IaC makes the duplication cheap.

Azure OpenAI only vs LiteLLM in front

- LiteLLM adds one hop, buys us multi-model routing, per-user quotas, and a clean seam for adding non-Azure models later if policy allows.

3 · Infrastructure as Code

Terraform, matching the existing ADO setup.

```
terraform/
├── modules/            # network, container-app, postgres, openai, langfuse, frontdoor
├── stacks/openwebui/   # the composed product
└── envs/
    └── <client>/
        ├── dev.tfvars
        ├── test.tfvars
        └── prod.tfvars
```

- **Remote state** in Azure Storage, one file per client/env, blob-lease locking
- **Naming + tagging** via `azurecaf` + Azure Policy; every resource traces to client/env/cost-centre
- **Guardrails** at subscription level (Defender for Cloud, Policy initiatives) — not re-implemented per stack
- Alternatives considered: **Bicep** (fine, but no reason to switch); **Pulumi** (overkill here)

4 · CI/CD — two pipelines, trunk-based

Infra pipeline (`terraform/**`)

1. PR → `fmt` · `validate` · `tflint` · `checkov` · `plan` as PR comment
2. Merge `main` → apply `dev` → approval → `test` → approval → `prod`
3. Same pipeline, per-client variable group, one ADO environment per client-env

App pipeline (`app/**`)

1. Build OpenWebUI container (pinned upstream + our config layer)
2. Push to **ACR** tagged with commit SHA, SBOM attached, **Trivy** scan gate
3. Deploy via **Container Apps revision** with traffic split (10 % → 100 %), auto-rollback on health-probe failure

Model changes as config

- `models.yaml` → PR → pipeline validates against AOAI deployments → pushes LiteLLM config
- **No image rebuild** for model additions/retirements

5 · Security & compliance

- **Private-only data plane.** No public endpoints on DB, OpenAI, Blob, Key Vault.
- **Entra ID SSO** into OpenWebUI; role mapping via groups.
- **Content Safety** on prompts + responses; prompt-injection guardrails enforced by Langfuse evals in CI.
- **Audit**: diagnostic settings → Log Analytics → optional SIEM export per client.
- **Data residency**: region pinned per client; CMK on Storage + Postgres when required.
- **Supply chain**: pinned upstream image digest, SBOM + vuln scan gate, signed commits.

6 · Observability & cost

Infra / app health

- Azure Monitor workbooks, alert rules as code, SLOs per service.

LLM layer — the one that actually matters

- **Langfuse** for traces, token cost, latency, eval datasets
- Prompt iteration becomes data-driven: compare model/prompt versions before promotion
- Eval suite runs in CI; failing evals block prod deploys

Cost

- Budget alerts per client subscription
- Per-user/-group token quotas in LiteLLM
- Container Apps scale-to-zero on non-prod

7 · Ways of working

- **Template repo** per product: `openwebui-azure-template`
  - New client = fork + tfvars + ADO variable group → first deploy in a day
- **Definition of done** per PR: plan clean · image scanned · dev deployed · Langfuse evals green
- **Model onboarding is a PR**, not a ticket — reviewer checks quota, cost, content-safety config
- **Runbooks live next to the code.** Postmortems drive module changes, not tribal knowledge.
- **Upstream tracking**: pin OpenWebUI version, test in dev weekly, promote monthly

8 · What I'd validate in week 1

- Client's existing Azure landing zone — plug into their hub or build standalone?
- Regulatory asks — CMK? data residency? log retention? DPA?
- Expected user count + model mix — drives Container Apps scale rules and AOAI quota requests
- SSO source of truth and group model

These answers change 2–3 cells in the trade-off table. Nothing below.

Thank you

Questions / deeper dive on any layer?

Architecture · IaC · CI/CD · Security · Observability · WoW